DESIGN AND STRUCTURAL ANALYSIS OF CONNECTING ROD USING ANSYS WORKBENCH

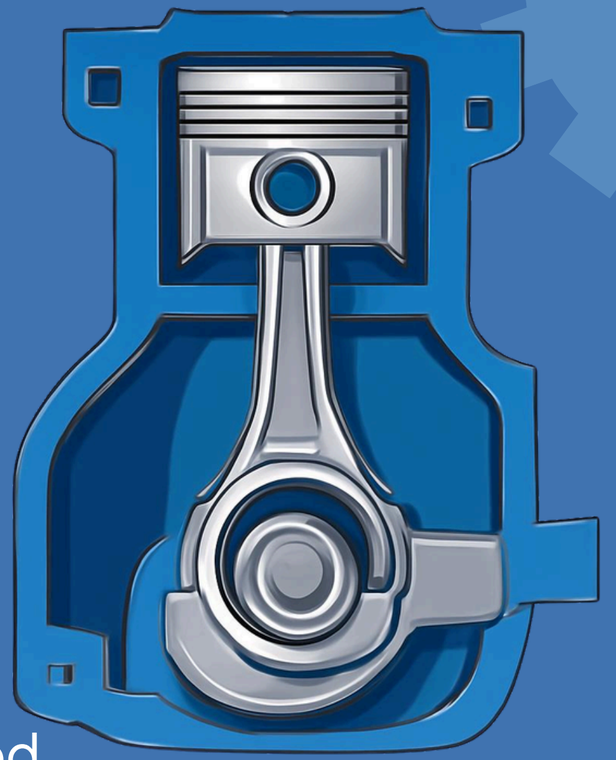




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SolidWorks 2025 & ANSYS Workbench 2025 R1

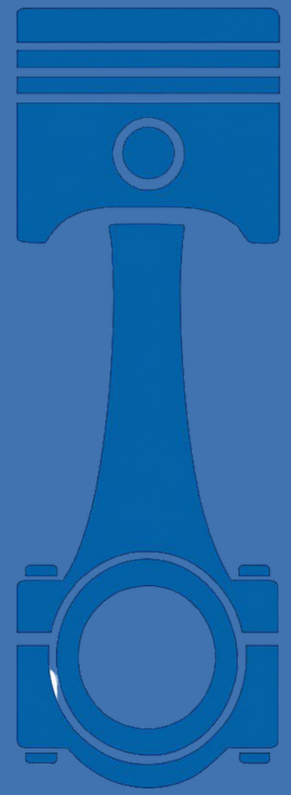
PROJECT OBJECTIVE

- Design a connecting rod using SolidWorks.
- Import the CAD model into ANSYS Workbench.
- Perform Static Structural Analysis.
- Determine Total Deformation.
- Determine Equivalent (Von-Mises) Stress.
- Evaluate the structural safety of the connecting rod.



PROBLEM STATEMENT

- Component: Connecting Rod
- Material: Structural Steel
- Analysis Type: Static Structural Analysis
- Purpose: Evaluate deformation and stress under applied load.
- Loading Condition: External force applied at the small end.
- Support Condition: Fixed support at the big end.



TOOLS & MATERIALS

Software
- SolidWorks 2025
- ANSYS Workbench 2026 R1

Material
- Structural Steel

Material Properties
- Young's Modulus: 200 GPa
- Poisson's Ratio: 0.30
- Density: 7850 kg/m^3
- Yield Strength: 250 MPa

SIMULATION WORKFLOW

1. Created the 3D connecting rod model in SolidWorks.

2. Imported the geometry into ANSYS Workbench.

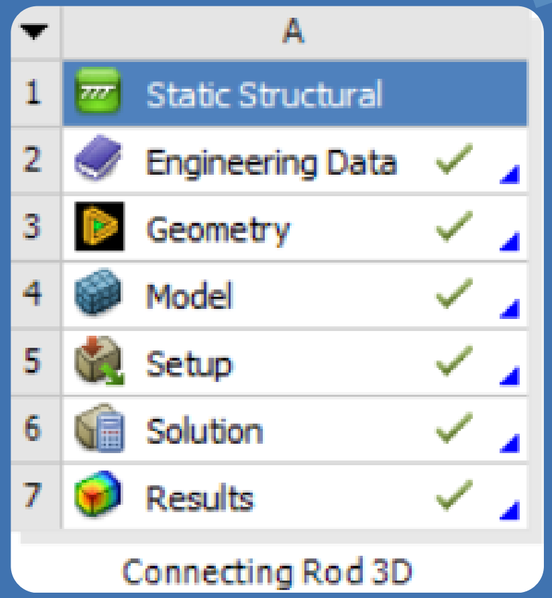
3. Assigned Structural Steel as the material.

4. Generated the finite element mesh.

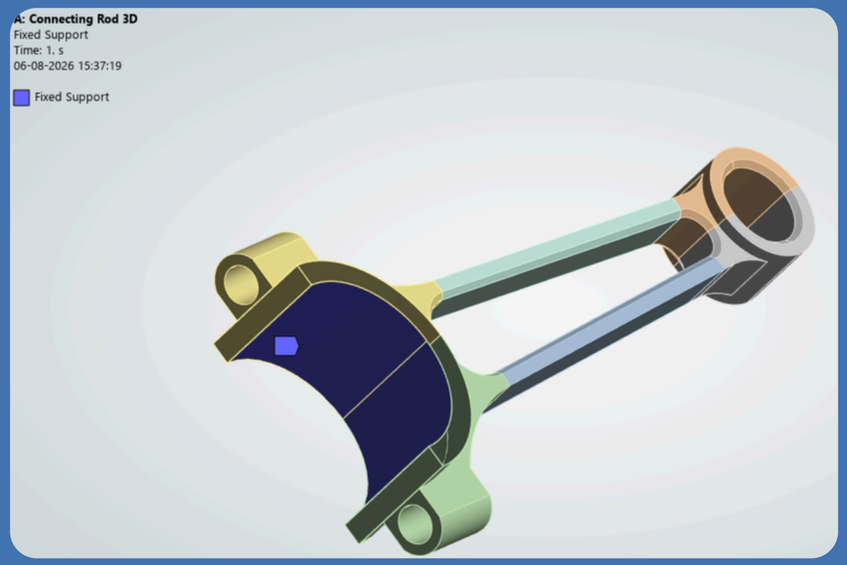
5. Applied Fixed Support and External Force.

6. Solved the Static Structural analysis.

7. Evaluated the Total Deformation, Equivalent Stress, and Safety Factor.

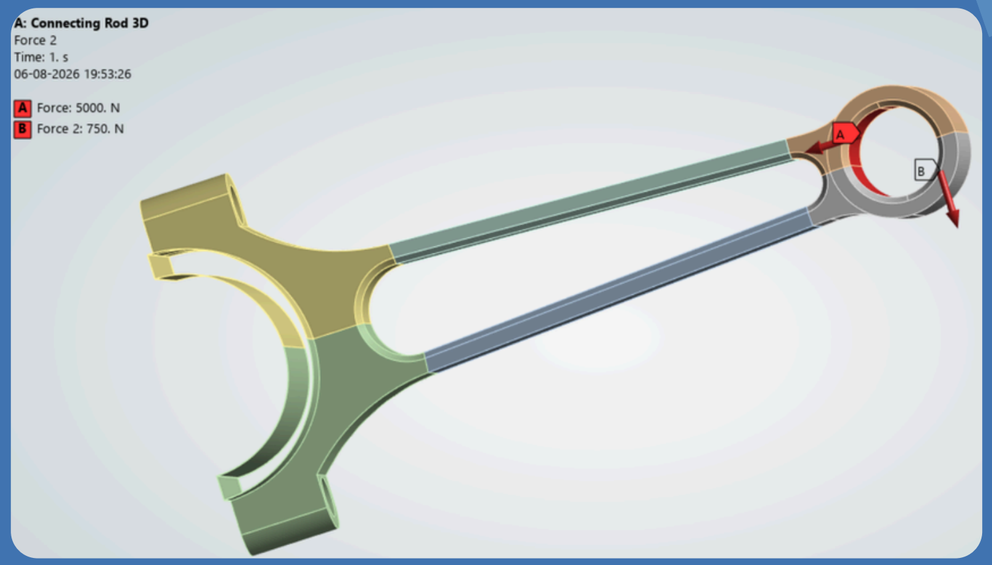


BOUNDARY CONDITIONS



Fixed Support

- Applied at the Big End of the connecting rod.



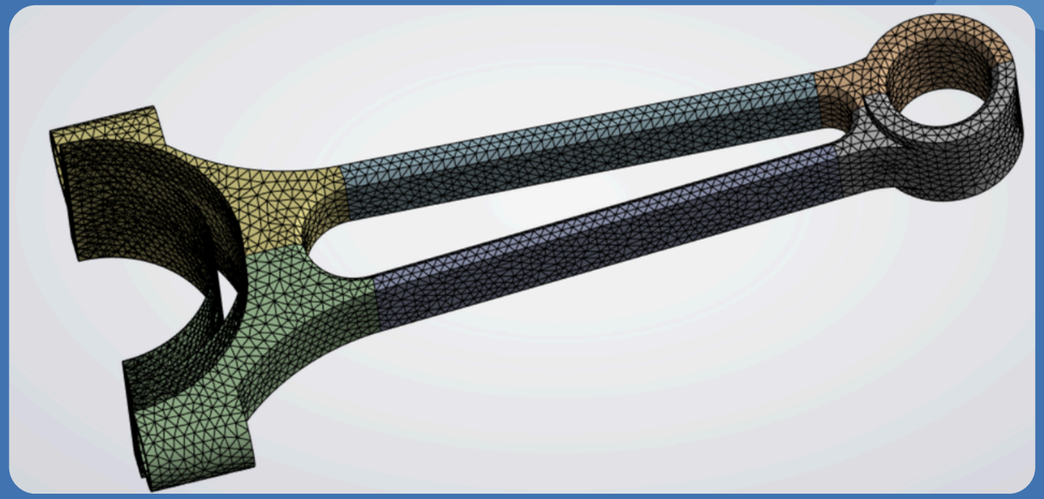
Applied Load

- **Force 1:** 5000 N
- **Force 2:** 750 N

MESH CONFIGURATION

Mesh Details:-

- Mesh Type: Automatic Tetrahedral Mesh
- Mesh Method: Patch Conforming
- Element Order: Program Controlled
- Mesh generated successfully for accurate structural analysis.



RESULTS OVERVIEW

SIMULATION RESULTS

Maximum Total Deformation
- 0.031969 cm

Maximum Equivalent (Von-Mises) Stress
- 1.4017×10^{8} dyne/cm^{2}

Minimum Safety Factor
- 15

OBSERVATION

• Maximum deformation occurred at the Small End of the connecting rod.

• Maximum stress was observed near the fillet region.

• The connecting rod is structurally safe under the applied loading conditions.

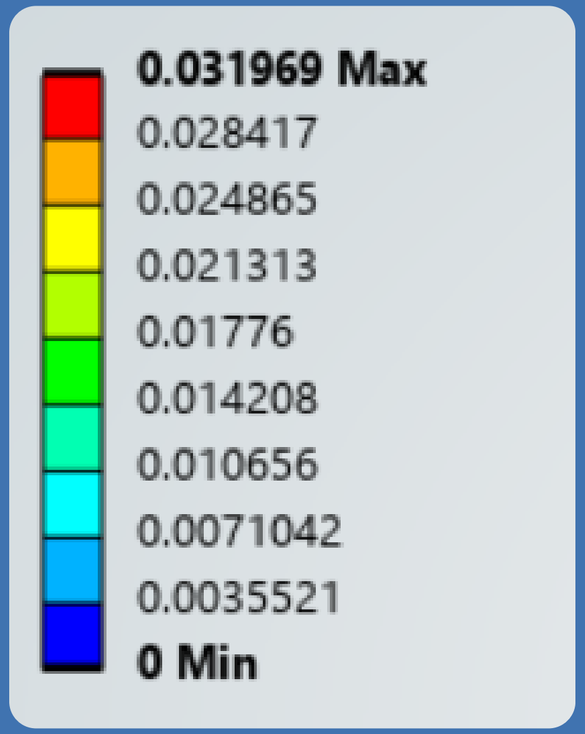


***Figure 8.1**: Total Deformation*

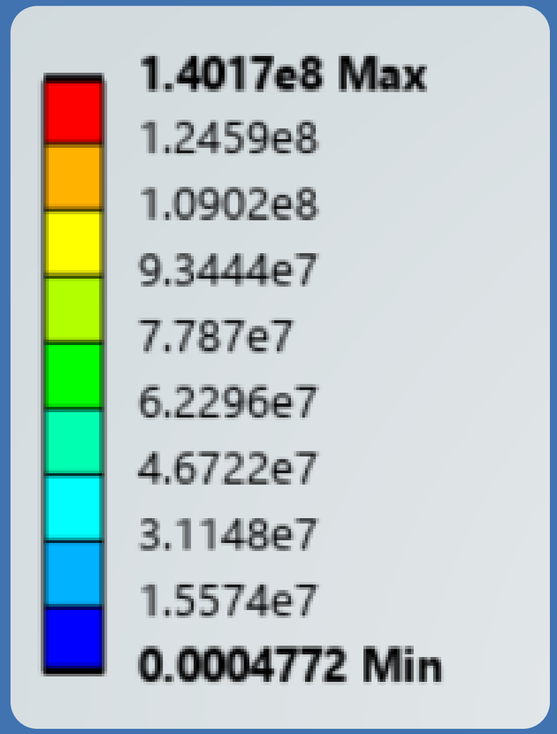


***Figure 8.2:** Equivalent (Von-Mises) Stress*

RESULT SNAPSHOTS

Simulation Result Images

- Total Deformation
- Equivalent (Von-Mises) Stress
- Safety Factor
- Finite Element Mesh

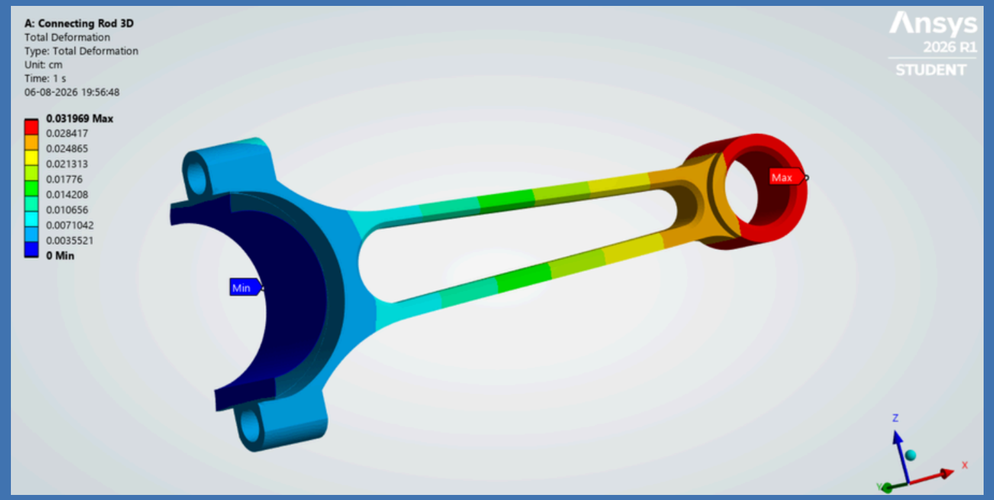


***Figure 9.1**: Total Deformation*

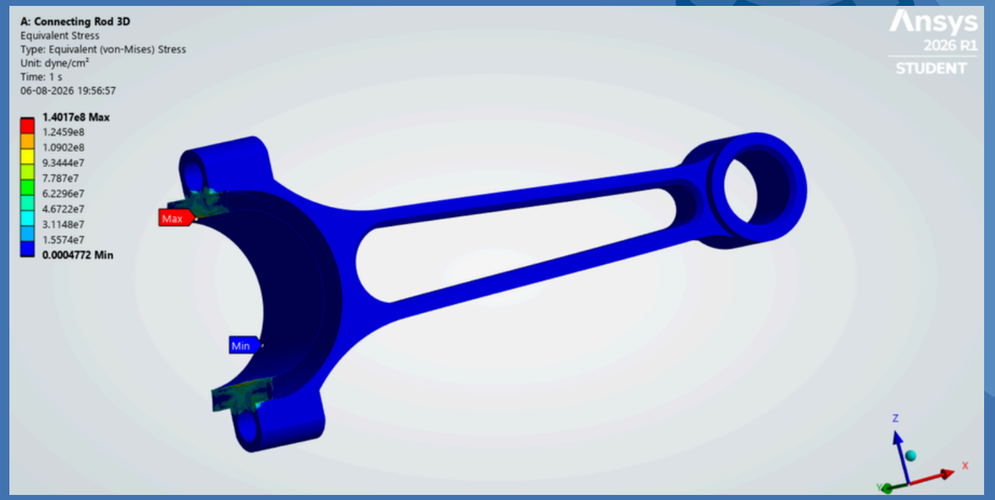


***Figure 9.2**: Equivalent Stress*

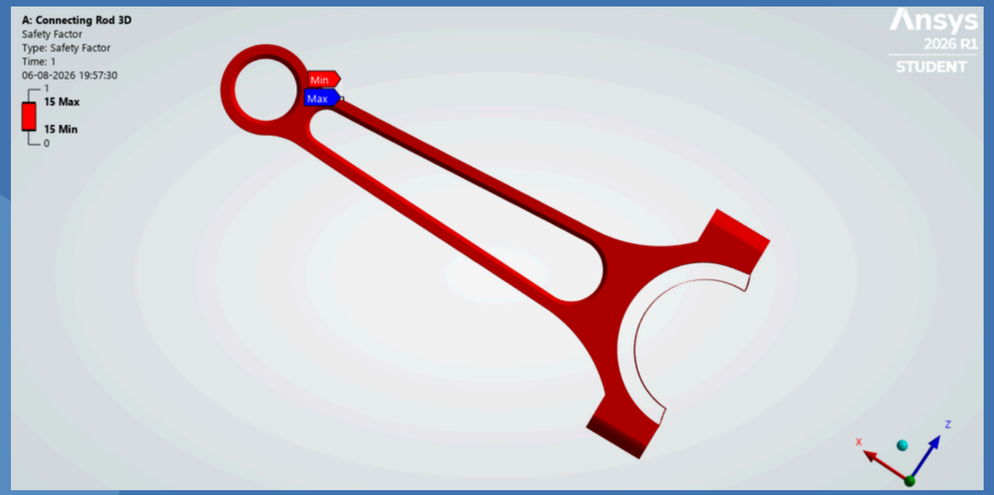


***Figure 9.3**: Safety Factor*

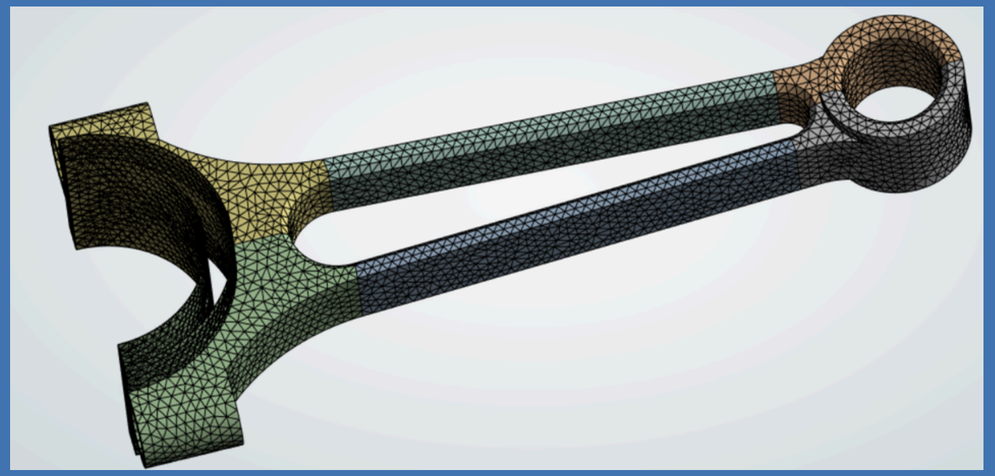


***Figure 9.4**: Finite Element Mesh*

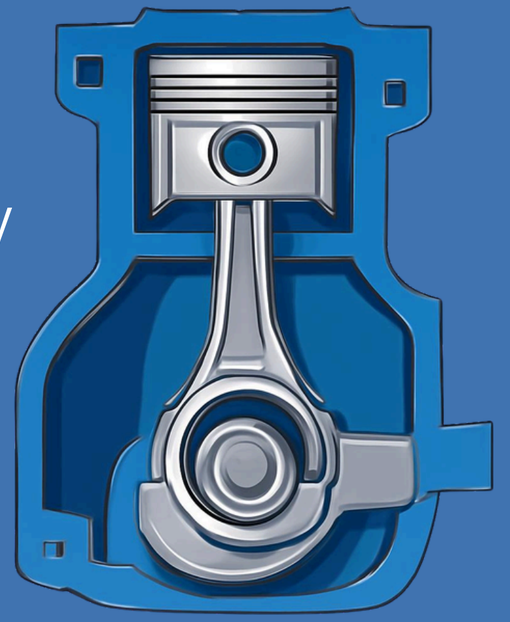
KEY LEARNINGS



- Learned 3D modelling of a connecting rod using SolidWorks.
- Performed Static Structural Analysis in ANSYS Workbench.
- Understood the importance of material properties and boundary conditions.
- Learned the process of mesh generation for FEA.
- Evaluated Total Deformation, Equivalent Stress, and Safety Factor.
- Gained practical knowledge of Finite Element Analysis (FEA).

SUMMARY & CONCLUSION

- Successfully designed the connecting rod using SolidWorks.
- Performed Static Structural Analysis in ANSYS Workbench.
- Evaluated Total Deformation, Equivalent (Von-Mises) Stress, and Safety Factor.
- The simulation results indicate that the connecting rod is structurally safe under the applied loading conditions.
- Finite Element Analysis (FEA) helps validate the design before manufacturing, reducing cost and improving reliability.



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